

Upper Assam Basin — NWIS Geological & Drilling Context

Synthetic research-summary document for retrieval, formation classification, and hazard-context demonstrations

IMPORTANT: This document is synthetic demonstration data prepared for the NWIS/SIH prototype. It must not be represented as an authentic confidential OIL record, real incident report, or verified SPE publication.

Purpose

This document supplies formation-level context so the NWIS assistant can connect a detected depth/formation to a known hazard class. It is deliberately labeled a synthetic research summary and is not an authentic SPE publication.

Formation Regimes

Formation	Approx. interval	Synthetic pore-pressure window	Fracture gradient	Primary hazard
Tipam	0–800 m	1.02–1.08 g/cc	1.65 g/cc	Washout / high permeability
Girujan	800–1500 m	1.10–1.15 g/cc	1.72 g/cc	Clay swelling / tight hole
Barail	1800–2500 m	1.18–1.25 g/cc	1.80 g/cc	Reactive shale / stuck pipe
Kopili	2500–3200 m	1.28–1.38 g/cc	1.88 g/cc	Overpressure / kick / influx
Sylhet	3000–3500 m	1.15–1.22 g/cc	1.52 g/cc	Fractured limestone / lost circulation

Geological / Drilling Features

Barail: the synthetic knowledge base represents reactive shale as a formation-specific drilling concern and links it to swelling, tight-hole behavior, torque increase, and pipe-entrapment cases. Kopili: the synthetic knowledge base represents selected intervals as potentially overpressured and links them to influx/gas-kick indicators. Sylhet: the synthetic knowledge base represents fractured/vuggy limestone and links it to severe mud-loss risk.

How This PDF Supports NWIS

This document does not replace well-specific reports. It provides background context that helps the retrieval layer classify a situation. For example, if DUL-104 is at 2205 m in Barail, NWIS can retrieve Barail-related context and then retrieve the more specific DUL-92 incident report.

RAG Chunking Suggestions

Chunk type	Recommended metadata
Formation	formation, depth_range, basin, hazard
Incident	well_id, event_type, depth, formation, date
Mitigation	well_id, event_type, action, outcome
Sensor pattern	well_id, formation, depth, parameter, anomaly_type

Prototype Data Governance

All synthetic documents should be marked as synthetic in the UI and metadata. Real deployment would require authenticated OIL historical reports, approved drilling procedures, actual well coordinates, validated formation tops, and real-time data interfaces. AI output should be decision support rather than an autonomous drilling command.